Problem Set for 26–May and 29–May

Exercise 0 Show that $U \otimes V \simeq V \otimes U$ with the following steps.

1. Construct $\varphi : U \otimes V \to V \otimes U$ via a 1:1 correspondence (refer to the preceding assignment). Describe φ by tracing the image of simple tensors.
2. Define $\psi : V \otimes U \to U \otimes V$ analogously.
3. Establish via 1:1 correspondence that both $\psi \circ \varphi$ and $\varphi \circ \psi$ are identical mappings.

> **Important**
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> In order to demonstrate that $\varphi : U_1 \otimes U_2 \to V$ is well-defined, the only permissible method is to identify a bilinear mapping $\Phi \in \mathrm{Hom}_{\mathbb{F}}(U_1, \mathrm{Hom}_{\mathbb{F}}(U_2, V))$ and define $\varphi(u_1 \otimes u_2) := \Phi(u_1)(u_2)$.
>
> Once φ has been defined, it is uniquely determined by the image of simple tensors.

Exercise 1 Show that the following is an isomorphism:

$$\mathrm{Hom}(U, V^*) \simeq \mathrm{Hom}(V, U^*), \quad [u \mapsto f_u] \mapsto [v \mapsto [u \mapsto f_u(v)]].$$

> **Warning**
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> U 和 V 未必有限维. 禁止取基, 禁止额外公理.

Exercise 2 Let V be a finite-dimensional vector space with basis $\{e_i\}$. Let $\{f_i\}$ denote the dual basis of V^*. Define the following mappings:

- $\Delta : \mathbb{F} \to V \otimes V^*, \quad 1 \mapsto \sum_{i=1}^{n} e_i \otimes f_i;$
- $\nabla : V \otimes V^* \to \mathbb{F}, \quad \sum u_i \otimes \varphi_i \mapsto \sum \varphi_i(u_i).$

For $f, g : V \to V$, determine the composition

$$\mathbb{F} \xrightarrow{\Delta} V \otimes V^* \xrightarrow{(f \otimes g^*)} V \otimes V^* \xrightarrow{\nabla} \mathbb{F}.$$

> **Tip**
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> The result must be a scalar: the determinant, or the trace, that is a question.

Exercise 3 Demonstrate that, for finite-dimensional vector spaces, there exists an isomorphism determined via simple tensors:

$$\Phi : V_1^* \otimes V_2^* \otimes V_3 \xrightarrow{\sim} \mathrm{Hom}(V_1 \otimes V_2, V_3), \quad f \otimes g \otimes x \mapsto [a \otimes b \mapsto f(a)g(b)x].$$

Now take $\otimes = \otimes_{\mathbb{R}}$ and $V_i = \mathbb{C}$ (the two-dimensional vector space with basis $\{1, i\}$). We know that the usual multiplication defines a map in $\mathrm{Hom}_{\mathbb{R}}(\mathbb{C} \otimes \mathbb{C}, \mathbb{C})$:

$$\times : \mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \to \mathbb{C}, \quad w \otimes z \mapsto w \cdot z.$$

What is $\Phi^{-1}(\times)$?

Exercise 4 (optional) Show that $\Phi^{-1}(\times)$ comprises a sum of no fewer than three simple tensors. Consequently, any multiplication algorithm for complex numbers necessitates at least three real multiplications.

> **Important**
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> Here, one multiplication step $\leftrightarrow$ one summand of simple tensors.

Exercise 5 Let $A \in \mathbb{C}^{m\times m}$ and $B \in \mathbb{C}^{n\times n}$ be normal matrices (i.e., $PP^H = P^H P$). Show that $AX = XB$ if and only if $A^H X = XB^H$.

> **Tip**
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> Try to kill this problem with tensor products in few lines.

Exercise 6 Let $A, B \in \mathbb{C}^{n\times n}$ be Xie-definite matrices (亚正定矩阵，见谢启鸿教材), i.e., $\mathrm{Re}(u^H \cdot A \cdot u) \geq 0$ with equality if and only if $u = \mathbf{0}$. Define the matrix C via component-wise multiplication (the stupid multiplication):

$$C = (c_{i,j}), \quad c_{i,j} = a_{i,j} \cdot b_{i,j}.$$

Show that C is also Xie-definite.

> **Tip**
>
> Try to kill this problem with tensor products in few lines.

Exercise 6 The tensor product space $U \otimes V$ (as defined in class) arises from the Cartesian product $U \times V$ along with a certain quotient construction. Explain why this quotient makes the resulting linear space larger, for instance:

$$\dim(\mathbb{F}^2 \times \mathbb{F}^3) = 5 < 6 = \dim(\mathbb{F}^2 \otimes \mathbb{F}^3).$$